Joint positivity, free Lucas coordinates, and commutative proof certificates

A tenth synthesis of explicit Diophantine consistency constructions

Research packet for the ProveIt repository

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Abstract

This iteration separates two kinds of progress. First, a nearest-square identity compresses two strict-positivity requirements over unrestricted integers into one. Applied to the pinned source, it gives a single polynomial in 28 integers of degree eight, and a companion in 26 integers of degree sixteen. A second completed reduction removes a high-degree Lucas-coordinate parameterization by proving that its usual congruence is redundant on the actual source slice. Second, an alternative proof source dispenses with the ordered stack: single-conclusion derivability is characterized by a finite nonnegative flow, then by products of pairwise coprime affine integers. Its local polynomial is explicit, but its global Diophantine compilation is not yet materialized or counted. An asymmetric three-Pell parity interface is also proved independently. Every final claim distinguishes a completed consistency equation from a local module or a proposed compiler. No global minimality, new Lean/Rocq certificate, or historical priority is claimed.

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1 Scope, source pin, and recommended reading

The target is an ordinary polynomial with fixed integer coefficients whose absence of solutions is provably equivalent, in ZFC, to Con(ZFC). In particular, no coefficient may depend on whether ZFC is consistent. Huge fixed integers are acceptable, but are part of the specification rather than hidden parameters. The artifacts include actual coefficient payloads, exact arithmetic circuits, mathematical proofs, source bindings, and independently authored finite checks.

The reviewed repository is `https://github.com/VladimirReshetnikov/ProveIt`, main commit `9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`. Eight newly integrated packages, source60 through source67, were reviewed as inert data. They were placed by commit `a2cd79b9120fc75f48dcbef6e38d0f8b18d64d49`. The detailed archive/member hashes and source paths appear in `review/SOURCE_PINS.json` and `PROVENANCE.json`. No program from an incoming archive was executed.

What "simplest" means here. There is no single scalar objective. We keep the unknown domain, number of unknowns, total degree, and binary arithmetic-operation count separate. A square is a multiplication in the circuit ledger; named registers are abbreviations, not additional quantified variables. An unrestricted integer count is not interchangeable with a positive or nonnegative count. Local witnesses in a bounded computation are not the variable count of the fully compiled equation.

Logical status. A completed equation below has a mathematical equivalence proof, using explicitly named inherited source-calibration and Pell theorems. The new finite audits establish algebraic identities, literal counts, payload integrity, and regressions. They do not discharge the universal theorems inside a proof-assistant kernel. Nor does an equivalence to consistency prove consistency itself.

2 Comparison of the eight incoming constructions

Package	Representative completed frontier	Main reusable idea
60: Positive Gap	9 positive, degree 1,797,796, 465 operations; 15 positive, degree24	Positive gap coordinates; cofinal padding; sharper carry bounds.
61: Congruence Carriers	13 positive, degree34, 93 operations; 37 integers, degree6; 34 integers, degree8	Move congruences into already existing carriers rather than add witnesses.
62: Sign and Norm	42 integers, degree4; 35 integers, degree6; 30 integers, degree8	Square-class separation and sign recovery; positivity guards suitable for joint compression.
63: Sign Recovery	42 integers, degree4; 14 positive, degree34, 87 operations	Remove explicit signs by arithmetic information in existing factors.
64: Structural Reductions	41 integers, degree4; 36 integers, degree6; 31 integers, degree8	Delayed square substitution and a product positivity guard.
65: Shared Taps	9 positive, degree 1,675,396, 458 operations; 10 integers, degree 7,428,120, 525 operations	Equal complete coefficient records share kernel positions; height 1873.
66: Previous iteration	9 positive, degree 1,845,940, 444 operations; 10 integers, degree 3,507,876, 494 operations	Gap congruences and shorter circuits; one-stack local proof source.
67: Oriented Guards	9 positive, degree 1,494,006, 487 operations; 10 integers, degree 3,327,556, 519 operations	Positivity-first orientation, signed square separation, suitable ten-integer finalizer.

These are selected tradeoffs, not a claim that each dominates every other version in its package. In particular the incoming best quartic remains 41 unrestricted integers and the incoming best sextic remains 35. Simply combining source64's elimination with source62's elimination cannot save another variable: the needed square row is absent in the 41-coordinate source. Counting that nonexistent row would be invalid.

The genuinely new complete reduction is the joint positivity lemma: it changes how inequalities are represented, not the coefficient placement or addition chain. The free-coordinate reduction is likewise structural: it removes a source-dependent divisibility obligation rather than merely encoding it more cheaply. The commutative proof architecture is a more radical change of source, but its final global cost is deliberately left unreported until compilation is completed.

New completed low-degree tradeoffs. The joint-positivity reduction takes the incoming 30-integer octic to 28 integers at the same degree. Nesting it gives 26 integers at degree16. These preserve all arithmetic source residuals and all four fixed ports. They do not lower the existing quartic or sextic records. The free-Lucas construction keeps ten unrestricted integers; its primary gain is degree, not a claim to beat every incoming operation-count record. The archived literal circuits provide the exact source-specific operation accounting.

New completed equation	Integer unknowns	Exact degree	Live operations
Joint positivity	28	8	residual-system specification
Nested joint positivity	26	16	residual-system specification
Free Lucas, shared-gap source	10	1,576,864	481

The first two are explicitly given sums of squared residuals; no operation count is assigned to an arbitrary expansion or evaluation order.

3 A nearest-square guard: 28 unrestricted integers at degree eight

The two independent positivity guards can be compressed without changing any Pell equation, compiler coefficient, or arithmetic register. For integer x, y, put

$$G(x, y, t) = xy - (t^2 - x - y + 1)^2.$$

Then $x, y > 0$ if and only if $G(x, y, t) > 0$ for some integer t. Indeed, $G > 0$ forces $xy > 0$, while $x = -a, y = -b$ with $a, b > 0$ would give $(t^2+a+b+1)^2 > ab$. Conversely, let $N = x+y-1$. Then $xy \geq N \geq 1$. Writing $m = \lfloor \sqrt{N} \rfloor$, choose $t = m$ if $N - m^2 \leq m$ and $t = m + 1$ otherwise. The square error has absolute value at most m, and its square is strictly less than N (the exact-square case has error zero). Thus $G > 0$.

Every positive integer is $\mathrm{Pos}(p, q, r) = 1 + p^2 + q^2 + r(r + 1)$ by the three-square theorem applied to $4n - 3$. Therefore the single residual

$$xy - (t^2 - x - y + 1)^2 - 1 - p^2 - q^2 - r(r + 1)$$

replaces two separate positive ternary forms, using four private integer coordinates rather than six. In the reviewed 30-integer octic, take $x = F\alpha$ and $y = F(\beta J + 1 - F - \alpha)$. Both are quadratic, so the new residual is quartic. All other residuals remain unchanged, including the guard on $\eta\zeta$. The resulting sum of 14 squared residuals has 28 unrestricted integer variables and degree exactly eight; the unchanged coefficient $[a^4 f^4] = 1$ certifies exactness.

The equivalence is existential, not a bijection of witness tuples. From an old zero choose the nearest-square coordinate and a positive ternary representation of G. From a new zero restore independent positive ternary representations of x and y. Composing these maps with the pinned source theorem gives the ZFC-provable equivalence between absence of integer zeros and Con(ZFC). No global minimality, external priority, or new proof-assistant certificate is asserted.

A second application gives a higher-degree companion. With $z = \eta(1 - V + jc + hE - \eta)$ and $g = G(x, y, t)$, replace all three old positive ternary guards by

$$gz - (u^2 - g - z + 1)^2 = \mathrm{Pos}(p, q, r).$$

Applying the proved lemma twice shows that this is equivalent to $x, y, z > 0$. Nine private coordinates become five, leaving 26 integers and 13 residuals. The new residual has coefficient -1 at t^8, so its square has coefficient 1 at t^{16}; the other residuals do not involve t. The exact total degree is therefore sixteen.

A general joint-positivity construction. For any fixed $n \geq 1$, put $P_1 = x_1$ and recursively

$$P_{k+1} = P_k x_{k+1} - (t_k^2 - P_k - x_{k+1} + 1)^2.$$

Repeated application of the two-input lemma gives

$$(x_1 > 0 \wedge \cdots \wedge x_n > 0) \iff \exists t_1, \ldots, t_{n-1} \in \mathbb{Z} \quad P_n > 0.$$

A final equation $P_n = \mathrm{Pos}(p, q, s)$ therefore uses only $n + 2$ private integer witnesses instead of $3n$. The price is increasing degree, potentially doubling at each nesting. The 28- and 26-integer constructions are the two- and three-input instances. This is a reusable consequence of the proved lemma, not a global optimality statement. The total $2n + 2$ integer coordinates for n original positive coordinates is consistent with the classical conversion attributed to Sun (1992) in Sun's 2026 introduction. The claim here is the explicit nearest-square application to these existing private guards, especially preservation of degree eight in the 28-variable construction, not discovery of that general variable-count conversion.

4 A fully specified low-degree equation

For convenient paper use, the following is a compact presentation of the 28-integer polynomial, equivalent term by term to the literal JSON. All letters in the list

$$J, F, \alpha, z, f, h, r, v, \tau, \eta, y, \gamma, o, j, E, c, a, V, I, u, \rho, t, p, q, s, p_3, q_3, s_3$$

are unrestricted integer unknowns. There are 28. Define abbreviations

$$Q = J\beta, \quad R = 8\rho^2 + (3 + 4a)\gamma, \quad x = F\alpha, \quad x' = F(1 + Q - F - \alpha),$$

$$x'' = \eta(1 - V + jc + hE - \eta), \qquad \mathrm{Pos}(p, q, s) = 1 + p^2 + q^2 + s(s + 1).$$

The following twelve residuals are retained from the source:

$$
\begin{aligned}
R_1 &= 8\rho^2 - 4u^2(1+Q),\\
R_2 &= a - E - (1+2v)(1+Q)^3,\\
R_3 &= E + 8\rho^2(E-a),\\
R_4 &= c - \eta + (a-E)(V - 1 - jc - hE),\\
R_5 &= \tau^2 - 1 - E(c-\eta)(1 - V + jc + hE) - E^2(c-\eta)^2,\\
R_6 &= R^2 + 2caR - (3+4a)c^2 - 1,\\
R_7 &= -V + jc - JM_C - JM_F(1+Q)\\
&\quad + (Q+2)(FQ^2 - \alpha Q - Q^2(Q+1)),\\
R_8 &= (1 - \alpha - F + Q)(K_0 + 4u^2) - F - Qz,\\
R_9 &= V + c - fo,\\
R_{10} &= I - (1+2r)c^2,\\
R_{11} &= I^2 + (1-f^2)(a^2 + 4a + 3),\\
R_{12} &= -1 + V^2I^2 + y^2 - y^2I^2.
\end{aligned}
$$

Add two residuals

$$R_{13} = xx' - (t^2 - x - x' + 1)^2 - \mathrm{Pos}(p,q,s), \qquad R_{14} = x'' - \mathrm{Pos}(p_3,q_3,s_3).$$

The promised single ordinary equation is

$$\boxed{P_{28} = \sum_{i=1}^{14} R_i^2 = 0.}$$

The letter v here is the JSON variable `v_scale`; the four new nearest-square/positivity variables are `pair_root`, `pair_pos1`, `pair_pos2`, `pair_pos3`. The last three are `EZ_pos1`, `EZ_pos2`, `EZ_pos3`. Thus there is no implicit extra coordinate in this compact notation.

Fixed coefficients. The four constants are the whole authenticated source34 tuple:

$$\beta = 2^{1064921} - 1, \qquad K_0 = K, \quad M_C = MC, \quad M_F = MF_{\mathrm{source}}.$$

They are fixed integers, supplied as minimal unsigned little-endian payloads under `integer_system_audit/coefficients/`. The last port is already shifted; replacing it by the native unshifted mask is invalid. The binding gives hashes, bit lengths, the 943 ordered tiles, and the explicit bounded coefficient recipe. The independently authored checker reconstructs all four integers from that recipe and compares them with the materialized bytes. The total raw payload is 532,479 bytes, or 10,466 bytes in the original compressed copies.

The 26-integer companion. Keep the first 21 variables and $R_1, \dots, R_{12}$. Introduce only t, t', p, q, s and set $g = xx' - (t^2 - x - x' + 1)^2$. The equation is

$$\boxed{P_{26} = \sum_{i=1}^{12} R_i^2 + \big(gx'' - (t'^2 - g - x'' + 1)^2 - \mathrm{Pos}(p,q,s)\big)^2 = 0.}$$

Its exact degree is sixteen. In ZFC one can prove, for each of these fixed polynomials,

$$\neg\exists \mathbf{x} \in \mathbb{Z}^n \; P_n(\mathbf{x}) = 0 \quad \longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

The new proof is the explicit existential reduction to source30, not a claim that all of source30's inherited calibration has been re-proved in this packet. That dependence is made inspectable by retaining the source30/Q42X residual map, the sign-recovery proofs, compiler proof, and source34 coefficient evidence. Transitive proof-assistant imports remain references to the pinned repository.

5 A free Lucas coordinate: removing the high-degree index carrier

The central new construction has ten unrestricted integer unknowns. Its literal source-specific circuit has exact degree 1 576 864 and 481 live binary arithmetic operations. The mathematical change is not a better addition chain or a different source layout. We prove that a high-degree defining congruence in the Lucas certificate is unnecessary on a fixed residue slice. Its former quotient coordinate becomes a direct, low-degree Lucas coordinate. The degree formula changes from $196a + 260$ to $96a + 160$, where $a = \deg A$.

All statements here concern ordinary mathematics formalizable in ZFC. The native compiler/refutation equivalence, Pell classification and the identified Lucas index lemmas remain explicit dependencies. This is not a new Lean or Rocq certificate, a consistency decision, or a global minimality claim.

5.1 The literal source prefix

The selected positive-gap shared-tap certificate has $W = 116$, $H = 1873$, 31 weights w_i, 22 targets q_j, and three kernel polynomials D_j. The complete finite lists are in `norm/final10/data/`; their coefficient-extraction identities are proved in the source section below. Put $\kappa = 2134940$, $b = 6t+4$, $g = bg_0$, and define

$$B = 2^\kappa b^2, \quad \ell = \sum_{i=1}^{31} B^{w_i}, \quad U_c = \ell + g, \quad G_c = D_2(B)U_c^2 + D_1(B)U_c + D_0(B),$$

$$A_T = \sum_{j=1}^{22} B^{q_j}, \quad N_0 = B^{117}, \quad N = B^{1991},$$

$$\mathrm{mask} = N_0 - 1 - (b-1)\ell, \quad J_c = 2G_c + A_T,$$

$$S_c = g + N_0 J_c, \quad T_c = \mathrm{mask} + N_0(B-2)A_T - 2,$$

$$X = N(S_c + T_c + 1) - S_c - 1,$$

$$Y = 2^{16\kappa - 23} b B^{2114}, \qquad \mathcal{C} = bB^{116}.$$

Every displayed sum has a fixed finite number of terms, and every exponent is a fixed integer. The whole native four-port tuple and the twelve materialized circuit leaves are bound independently; they must not be substituted by the different source34 tuple used in P_{28} and P_{26}. The new degree bounds are $\deg X = 7962$, $\deg Y = 4229$ and $\deg A = 16424$. The independently checked primary circuit uses 301 multiplications, 155 additions and 25 subtractions, totaling 481. Compared with the incoming ten-integer degree-3,327,556/519-operation construction, both metrics decrease. It also improves the previous iteration's 494-operation, degree-3,507,876 variant. The new literal circuit treats the materialized threshold integer $2^{34159017}$ as one fixed leaf. Retaining the predecessor's five constant-building gates would give 486 operations, not 481. Both counts are reported to separate structural simplification from constant folding; neither is a bit-complexity or execution-time estimate.

5.2 The complete finalizer

Take the fixed source polynomials $X, Y, \mathcal{C}$ and its normalizations

$$f_{\mathrm{old}} = 2t + 1, \qquad b = 6t + 4, \qquad g = bg_0, \qquad k = b(3k_0 + 2).$$

The exact source prefix and all four coefficient ports are specified in the accompanying source section and literal data. Its universally valid congruences are

$$X \equiv 1 \pmod 2, \quad X \equiv 0 \pmod 3, \quad b \mid X+1, \quad b \mid Y.$$

The ten supplied variables are $(t, g_0, c_0, k_0, l, w, x, y, z, m) \in \mathbb{Z}^{10}$. Every other symbol below is a polynomial abbreviation. Define

$$\begin{gathered}
L = lY, \quad U = 2LX, \quad V = 4gwY, \quad W = bw, \\
A_{1/2} = (LX)(V+1), \quad A = 2A_{1/2}, \quad B_1 = 2X+1, \quad \boxed{C = 6c_0 - 1}, \\
P_{1/2} = (LX)UV, \quad P = 2P_{1/2}, \quad K = X + 1 + k(P-2), \\
S = 2A - 5, \quad T = 3WC - 2(W^2-1), \quad Q = (P_{1/2}^2 - 1)K^2 + 1, \\
\Delta = 8\mathcal{C}^3 g K^2 - g^2(32(C-KL)^2\mathcal{C}^3 + g^2K^2), \\
\xi = zx, \quad D = (A^2-4)C^2 + 4, \quad E = C^2 D\xi, \\
F = 4(A^2-4)E^2 + 1, \quad G = 2(A+2)(A-2)^2E^2 - 1, \\
\Omega = 16E, \quad H = C + F(B_1 - C + \Omega y), \quad I = (G^2-1)H^2 + 1.
\end{gathered}$$

The circuit uses the identical expression $G = (A_{1/2} - 1)(F-1) - 1$ to share operations. The subscripts $1/2$ name explicit polynomial registers, not divisions in the equation.

Keep the original one-modulus intersective finalizer:

$$\begin{gathered}
\rho = T + Sy, \qquad \Sigma = \xi S^5, \\
\Theta = (\rho^3 - \rho S^2 - S^3)(\rho^2 + 23S^2), \\
r = bQ, \qquad \delta = 9\Delta - 3, \qquad a_0 = r + \delta(rz)^2, \quad a_1 = DFI, \quad Z = \Sigma m - \Theta.
\end{gathered}$$

The single equation is

$$\boxed{\left(Z^2 - \Sigma^2(a_0 + a_1)\right)^2 - 4\Sigma^4 a_0 a_1 = 0.} \tag{1}$$

Every exponent is fixed. There are no hidden quotient, root, binomial, variable-exponent, or bounded-universal operations. All fixed leaves are materialized integers with hashes. In particular the new parameterization has no requirement $(A-2) \mid C - B_1$ built into it.

5.3 A strengthened signed Lucas certificate

Lemma. Suppose $A, B, C, \xi, y \in \mathbb{Z}$ satisfy

$$6 \mid A, \quad B \equiv 3 \pmod 4, \quad 1 < B < |A|/2 - 1, \quad C \equiv 5 \pmod 6, \quad \xi \neq 0.$$

Use D, E, F, G, Ω, H, I as above with B in place of B_1. If DFI is a square, then $C = u_B(A)$, where $u_0 = 0, u_1 = 1, u_{n+2} = Au_{n+1} - u_n$. No hypothesis $(A-2) \mid C - B$ is needed.

Proof. Here C is odd and nonzero. Since $|A| > 2$, the three integers D, F, I are positive. The certificate identities

$$F \equiv I \equiv 1 \pmod D, \qquad 4I \equiv D \pmod F$$

and oddness of F make them pairwise coprime, so they are separately squares. The usual Pell classification gives integers s, μ, t with

$$C = u_s(A), \qquad 4E = u_\mu(A), \qquad H = u_t(2G).$$

Put $n = |\mu| > 3$. The parity identities show that n is even and s, t are odd. The Lucas index-separation lemma applied to the congruence modulo F gives $s \equiv \pm t \pmod{2n}$. The rank-divisibility lemma applied to $C^2 \mid u_n(A)$ gives $C \mid n$. Finally $H \equiv B \pmod{\Omega}$ and $2G \equiv -2 \pmod{\Omega}$ give $t \equiv B \pmod{\Omega}$. Thus

$$s \equiv B \pmod{2C} \quad \text{or} \quad s \equiv -B \pmod{2C}. \tag{2}$$

These are precisely the congruence-free classification and index steps of the modified certificate; the new work is to remove its exceptional ranks and sign without the old defining congruence.

For every $j \geq 1$, direct paired induction in the recurrence proves

$$u_{2j}(A) \equiv (-1)^{j-1} jA \pmod{A^2}, \qquad u_{2j+1}(A) \equiv (-1)^j \pmod{A^2}. \tag{3}$$

The base is $u_2 = A, u_3 = A^2 - 1$. The next two recurrence steps multiply the second congruence by A and subtract the first, then multiply again by A and subtract the second. This works for negative A as well.

If $s = \pm 1$, then $C = \pm 1$ and $D = A^2$. Therefore $A^2 \mid u_n(A)$. Writing $n = 2j$, the first congruence in (3) yields $A \mid j$. Consequently $2|A|$ divides both $2n$ and $\Omega = 16A^2\xi$. Combining $s \equiv \pm t \pmod{2n}$ with $t \equiv B \pmod{\Omega}$ forces $B \equiv \pm 1 \pmod{2|A|}$, contradicting $1 < B < |A|/2 - 1$. This argument does not assume the generally false divisibility $n \mid u_n(A)$.

Hence $|s| \geq 2$. Lucas growth gives $|C| = u_{|s|}(|A|) \geq |A| > B$ and $|s| \leq |C|$. The two congruences in (2) therefore force $s = B$ or $s = -B$: a nonzero multiple of $2C$ would be too large. Since $B \equiv 3 \pmod 4$, the second congruence in (3) gives $u_B(A) \equiv -1 \pmod{A^2}$, and thus modulo3. The branch $s = -B$ would give $C \equiv 1 \pmod 3$, contrary to $C \equiv 5 \pmod 6$. Therefore $s = B$, as required. □

The odd, nonzero residue slice is essential: $A = 12$, $B = 3$, $C = 0$, $\xi = 1$, $y = 0$ gives $DFI = 4$, although $u_3(12) = 143$.

A reusable unsigned interface. The mechanism is not restricted to this particular circuit. For even A, odd C, odd B with $1 < B < |A|/2 - 1$, and nonzero ξ, the same square-certificate index argument gives $C = \pm u_B(A)$ without assuming $A - 2 \mid C - B$. To handle both exceptional indices $s = \pm 1$, use $C^2 = 1$: then $D = A^2$, $4E = 4A^2\xi$ and the paired recurrence forces $2|A| \mid n$, just as above. Both exceptions contradict the range of B. For $|s| \geq 3$, the growth and index congruences give $s = \pm B$. The fixed residue conditions $3 \mid A$, $B \equiv 3 \pmod 4$, $C \equiv -1 \pmod 3$ then select the positive index. Thus the primary construction is a corollary of a stronger arithmetic interface, not merely a shorter evaluation DAG. Its dependency on the standard Lucas index comparison is unchanged; no claim of literature priority is made.

5.4 Signed soundness of the ordinary equation

The source and k normalization imply $b \mid K$, $3 \mid K$, and

$$Q \equiv 1 \pmod{b^2}, \qquad Q \equiv 1 \pmod 3, \qquad \gcd(b, Q) = 1.$$

On every signed assignment $3 \mid A$, $C \equiv -1 \pmod 3$, so $D \equiv 0$ modulo3 and $a_1 \equiv 0 \pmod 3$. In contrast $r = bQ \equiv 1 \pmod 3$ and $\delta \equiv 0 \pmod 3$, so $a_0 \equiv 1 \pmod 3$. Thus $a_0 \neq a_1$ *unconditionally*, before any positivity argument.

The odd integer $S = 2A - 5$ is nonzero. The polynomial $\pi(v) = (v^3 - v - 1)(v^2 + 23)$ has no rational root; hence $\Theta = S^5\pi(\rho/S) \neq 0$. A zero of (1) therefore has $\Sigma \neq 0$. With $q = Z/\Sigma$, the equation gives $(q^2 - a_0 - a_1)^2 = 4a_0a_1$. Since $a_0 \neq a_1$, $q \neq 0$. The rationals

$$u = \frac{q^2 + a_0 - a_1}{2q}, \qquad v = q - u$$

have squares a_0, a_1. Rational-square-root integrality makes u, v integers. Thus a_0, a_1 are squares and $\Sigma \mid \Theta$. A prime-valuation comparison in the homogeneous quintic yields $S \mid T$: if $v_p(T) < v_p(S)$, the two factors of Θ have valuations $3v_p(T)$ and $2v_p(T)$, contradicting $S^5 \mid \Theta$.

The factorization $a_0 = r(1 + \delta r z^2)$ has two coprime nonzero factors. Since $r \equiv 1 \pmod 3$, the signed-square possibility makes r and the bracket positive squares. Applying the same argument to $r = bQ$ gives b, Q positive squares, hence $b \geq 4$. If $\Delta \leq 0$, then $\delta = 9\Delta - 3 \leq -3$, making the bracket negative. Thus $\Delta > 0$. Its defining inequality gives $K \neq 0$ and $0 < g < 2\mathcal{C}$. If $l = 0$ then $Q = 1 - K^2 \leq 0$, a contradiction. Hence $l \neq 0$; $f_{\text{old}} = 2t + 1$ is already nonzero.

The source bounds now give $X > 3b$ and $Y > 256$. Since $V + 1$ is odd and nonzero,

$$|A| = 2|l|XY|V + 1| \geq 2XY > 2B_1 + 2.$$

All hypotheses of the strengthened lemma hold: $6 \mid A$, $B_1 \equiv 3 \pmod 4$, $C \equiv 5 \pmod 6$, and $\xi \neq 0$. It gives $C = u_{B_1}(A)$. The recurrence modulo $A - 2$ then gives $(A - 2) \mid C - B_1$ *as a consequence*. Restore the proof-only integer $h = (C - B_1)/(A - 2)$ and apply the unchanged reverse binomial bridge. The calibrated source decodes a positive source zero and hence a finite ZFC refutation. No sign of an arbitrary supplied auxiliary variable was silently assumed.

5.5 Completeness

Start from a positive source zero and its canonical arithmetic witnesses. They have $C = u_{B_1}(A)$, $6 \mid A$, and $B_1 \equiv 3 \pmod 4$. Thus C is odd and $C \equiv -1 \pmod 3$, giving the integer $c_0 = (C+1)/6$. The old constructive congruences still supply k_0. Here b, Q are positive squares and $\Delta > 0$. The positive integer

$$\delta = 3(3\Delta - 1)$$

has 3-adic valuation exactly1, and is therefore nonsquare; the same holds for δr since $r = bQ$ is a positive square. Pell's theorem supplies positive u, z with $u^2 - \delta r z^2 = 1$, making $a_0 = ru^2$ square.

The modified Lucas interpolation theorem first supplies a positive ξ divisible by this z, and then permits every residue class of y modulo that fixed ξ while retaining DFI square. Put $x = \xi/z$. Since $S \mid T$, write $v_0 = T/S$. The quintic π has a root modulo every positive integer, so prescribe y with $\pi(v_0 + y) \equiv 0 \pmod \xi$. Then $\Sigma \mid \Theta$. Taking

$$m = \Theta/\Sigma + \sqrt{a_0} + \sqrt{a_1}$$

gives an integer solution of (1). Thus the construction is existence-equivalent to the fixed source, and negating the two existence statements gives the required equivalence with Con(ZFC).

5.6 Exact degree and verification boundary

Let $a = \deg A$. Since C now has degree1,

$$\begin{aligned} &\deg D = 2a + 2, && \deg E = 2a + 6, && \deg F = 6a + 12, \\ &\deg G = 7a + 12, && \deg H = 8a + 19, && \deg I = 30a + 62, \\ &\deg a_1 = 38a + 76, && \deg \Sigma = 5a + 2. \end{aligned}$$

For the included compatible source, the guard and center have strictly lower degree than the dominant $\Sigma^4 a_1^2$ term. Consequently

$$\deg F_{10} = 4(5a + 2) + 2(38a + 76) = 96a + 160.$$

Exact nonzero diagonal leading coefficients are checked with the actual fixed tuple modulo three primes. They certify that the upper degree is attained; they are not sampled estimates of an unproved upper bound. Separate checks authenticate coefficient bytes, exact source extraction, full modular formulas, signed residue separation and live circuit topology. Finite checks do not replace the unbounded mathematical proof or its stated arithmetic/compiler dependencies.

6 A compatible positive-gap shared-tap source

This auxiliary improvement combines the source65 shared coefficient taps with the round9 positive-gap coordinate substitution. It does not change the four native coefficients or claim a new universal-polynomial theorem. All31 source variables range over $\mathbb{Z}_{>0}$. The substitutions $P_{\mathrm{old}} = P + \eta + \zeta$ and $N_{\mathrm{old}} = N + 1$ are bijections on the positive zero sets: the original equations imply $P_{\mathrm{old}} - \eta - \zeta = E(c - \eta) > 0$ and $N_{\mathrm{old}} = q^3 \geq 8$. The original31 coefficient tuple is retained together and materialized.

Let w_i be the31 distinct positive weights, q_j the22 targets and $U(T, x) = \sum_i x_i T^{w_i}$. The serialized certificates prove the formal identities

$$[T^{q_j}](D_2(T)U(T,x)^2 + D_1(T)U(T,x) + D_0(T)) = 2\rho_j(x) \quad (1 \leq j \leq 22).$$

Equal complete coefficient records may share one kernel position; they are not added together. Exhaustive symbolic extraction checks every monomial, including the absent ones. The primary source uses86 records and81 unique positions, compared with source65's89 records and84 positions. Its layer degrees are $(1384, 1732, 1641)$, with $W = 116$, $Q = 1841$ and $H = 1873$. Thus the packing removes three positions at unchanged height. The85-record alternative does not satisfy this layout and is not substituted silently for the primary source.

The native kernel coefficient bound remains $M = 2^{2134927}$, checked on the actual fixed integers. For $B \geq 4096Mb^2$ and $U = \sum_i B^{w_i} + g \geq B^{116}$, the quadratic leading coefficient is1. Its negative tail is less than half its leading term. The possible negative linear and constant contributions, relative to $B^{1641}U^2$, are at most $2M/B^{25}$ and $2M/B^{489}$. Consequently the source convolution is positive before digit recovery. The margin $H - Q = 32$ preserves the existing carry bounds. Every weight is positive, so canonical encoded witnesses remain divisible by B, and in particular by b^2.

The artifact auditor verifies110 exact target identities across five certificates, rejects five coefficient mutations, validates all four compressed and raw native coefficient hashes, and establishes that every materialized kernel coefficient has absolute value below$M$. These checks are finite algebraic certificates, not a new kernel-checked proof of the inherited source-to-consistency theorem.

7 Disjunctive scratch-space reuse

This section gives a fully explicit improvement to the *local* proof transition relation. It does not claim that a complete consistency equation with the following variable counts has been constructed. The remaining bounded-universal elimination is a separate cost.

The elementary principle. Suppose finitely many relations have presentations

$$R_j(\mathbf{x}) \iff \exists \mathbf{z}_j \in \mathbb{N}^{m_j} \quad Q_j(\mathbf{x}, \mathbf{z}_j) = 0.$$

Put $m = \max_j m_j$. Then

$$\bigvee_j R_j(\mathbf{x}) \iff \exists \mathbf{z} \in \mathbb{N}^m \quad \prod_j Q_j(\mathbf{x}, z_1, \ldots, z_{m_j}) = 0.$$

Indeed a product of integers is zero exactly when a factor is zero. Only one branch's scratch values need be correct; unselected branches impose no obligations. In contrast, flattening the whole arithmetic circuit before taking its zero set generally forces scratch equations for every branch and pays a sum rather than a maximum.

Application to the one-stack source. Use $I(a,b) = 2(a+b)(a+b+1)+4b$, with membership, universal quantification, and equality obtained by adding $1, 2, 3$ respectively. Let $L(s,h,t) = 2s - 2 - (h+t)(h+t+1) - 2t$. The equation $L(s,h,t) = 0$ expresses a stack cell with head h and tail t. There are twelve alternatives: truth push, eight axiom-template pushes, the guarded seed-bundle push, modus ponens, and generalization. Each branch has its own arithmetic circuit, but the variable names $q_0, q_1, \ldots$ are reused across all branches. Each distinct constructor uses one nonnegative wire

$$q = (u+v)(u+v+1), \qquad I(u,v) = 2q + 4v.$$

The four shared parameters are a, b, c, d. In the eighth axiom d is the fourth schema parameter. In modus ponens it is the intermediate stack; in the bundle it certifies the guard. These roles need not agree between unselected alternatives.

For the first construction the bundle guard is

$$q_{26} = (c-a)^2, \quad q_{27} = (c-b)^2, \quad q_{26}q_{27} = (d+1)^2.$$

All residuals are quadratic. The twelve sums of their squares have degree four, giving a product of degree 48, with 28 common wires and four parameters: 32 local witnesses, compared with the previous 89. The exposed stack coordinates are not included in these counts.

The recommended local tradeoff. Instead use the literal guard

$$((c-a)(c-b))^2 - (d+1)^2 = 0.$$

Over natural parameters this holds for some d precisely when $c \neq a, b$. It saves the two square wires. Also substitute the defining polynomial of the bundle's outermost constructor wire into its stack-cell residual and omit that wire equation. The substitution is exact and does not alter existential solvability. The bundle sum of squares now has degree eight; the eleven remaining factors still have degree four. Thus the product has exact degree $8 + 11 \cdot 4 = 52$ and only $25 + 4 = 29$ local witnesses. No expansion of the twelve-factor product is necessary: the attached JSON lists every residual and each expanded branch sum of squares, and specifies that their product is the final local polynomial.

Correctness. If the polynomial vanishes, one branch sum of squares vanishes, and hence each of its residuals vanishes. By induction through that branch's constructor circuit its wires have their intended values. The stack residuals therefore state the corresponding legal proof transition. The bundle guard gives the required disequalities. Conversely, for any legal transition choose its schema parameters and assign the selected branch's wires their ordinary nonnegative values; choose arbitrary values for unused common wires. All selected residuals vanish. Eliminating an outermost wire preserves both directions because its value is uniquely determined and nonnegative. This proves equality of the local relations, not merely agreement on tested examples.

Limits and reproducibility. The JSON files are literal ordinary integer polynomials, with no hidden variable exponents or unbounded products: the outer product has exactly twelve factors. The global equivalence to $\neg\,\mathrm{Con}(\mathrm{ZFC})$ would follow by substituting this local relation in the established finite one-stack-run characterization and completing bounded-universal elimination. That global constructor is not executed here, so these local counts must not be compared directly with the existing nine- and ten-variable completed consistency equations. The independent checker reads the JSON without importing the generator, evaluates selected branches using a separate integer interpretation of the constructors, and rejects endpoint mutations. These tests supplement the algebraic argument; they are not kernel certificates.

Dependency-aware refinement: seventeen local witnesses. The previous paragraph can be improved without increasing the degree. Introduce a wire for the *whole* constructor output $z = I(u,v) = 2(u+v)(u+v+1)+4v$, rather than a wire only for $(u+v)(u+v+1)$. Whole-output wires prevent linear tails from propagating extra dependencies. The bundle now has 27 constructor wires. In their undirected dependency graph, the set

$$\{1,3,4,5,7,9,12,15,17,19,21,22,24,26\}$$

is independent. Substitute the defining quadratic of each selected wire and delete its defining residual. Since no selected wire depends on another selected wire, the substituted expressions have degree two, and each remaining residual has degree at most four. There are only 13 remaining wires, with the same four base parameters. The twelve-factor local transition polynomial therefore has 17 witnesses and degree 52. Its expanded bundle sum of squares has 1110 monomials. The global correctness argument is unchanged: this is exact existential elimination of nonnegative, uniquely defined scratch values.

For use in the proof-flow architecture, replace stack-cell residuals by $p = q = 0,\ v = \rho$ in each axiom branch, by $q = I(p,v)$ in the modus-ponens branch, and by $q = 0,\ v = \mathrm{All}(a,p)$ in the generalization branch. The truth branch is $p = q = v = 0$. The resulting literal polynomial has three exposed formula labels, the same 17 local witnesses, and degree 50: the truth factor now has degree two rather than four. The independent-wire elimination uses the same fourteen-index set.

8 A different source: commutative proofs and affine products

The main architectural idea in this component is to discard proof order. The following result explains why a cycle check is unnecessary. It is a mathematical theorem, not a claim of historical priority for flow certificates.

Single-conclusion flow theorem. Let F be a set of formula codes and R a family of finitary rules, each having a premise multiset P_r and exactly one conclusion c_r. Axioms have empty premise multisets. Write D for the least rule-closed subset of F, and e_f for the basis vector of f in $\mathbb{Z}^{(F)}$. Then

$$t \in D \quad \Longleftrightarrow \quad \exists (n_r) \in \mathbb{N}^{(R)} \quad \sum_r n_r \left(e_{c_r} - \sum_{p \in P_r} e_p \right) = e_t.$$

Completeness counts occurrences in a finite proof tree: every non-root occurrence cancels once. For soundness, suppose $t \notin D$, and sum the coordinates outside D. Each rule contributes at most zero: an underivable conclusion must have an underivable premise. The right-hand side contributes one, a contradiction. This argument even proves soundness for nonnegative rational multiplicities and a coordinatewise lower bound by e_t. A finite certificate can be converted constructively into a proof DAG by saturating its finite set of selected rules.

The single-conclusion restriction is essential. The rule with premise a and two conclusions a, t has balance e_t, without deriving either formula from no axioms. Thus the result is not a general Petri-net reachability criterion. Nonnegative multiplicities are likewise essential.

Free truth and uniform arity. In the calibrated C222 calculus, truth code 0 is derivable. We may therefore ignore coordinate 0, pad every axiom with two zero premises, and pad generalization with one zero premise. Every rule now has the form $(p,q) \longrightarrow v$. The target is code 1. The inherited C222-to-ZF calibration, followed by the constructible-choice bridge from ZF to ZFC, shows in ZFC that existence of such a finite flow certificate is equivalent to $\neg\,\mathrm{Con}(\mathrm{ZFC})$.

Coprime affine weights. Choose $m \geq 1$ bounding every label in a proposed certificate, and a positive common multiple M of $1, \ldots, m$. Set

$$f(j) = 1 + jM \qquad (0 \leq j \leq m).$$

Then $f(0) = 1$, and the positive-label weights are pairwise coprime. Indeed, a common divisor of $1 + iM$ and $1 + jM$ is coprime to M and divides $j - i$; the latter divides M, forcing that common divisor to be one. Unique factorization consequently gives the exact equivalence

$$\prod_{i<N} (1 + v_i M) = (1 + M) \prod_{i<N} (1 + p_i M)(1 + q_i M)$$

if and only if the uniform rules have balance e_1 modulo coordinate 0. No primality predicate and no exponentiation by a formula code appears. Taking $M = m!$ proves existence; factorial is not being treated as a polynomial. Merely taking an arbitrary large M is not a substitute for the coprimality condition.

A one-accumulator source. A product certificate can be realized by a path

$$S_0 = 1 + M, \qquad S_N = 1, \qquad S_{i+1}(1 + v_i M) = S_i(1 + p_i M)(1 + q_i M).$$

Multiplying these equations and cancelling the positive internal accumulators gives the product identity. Positivity follows inductively from $S_0 > 0$ and positivity of every factor. Conversely, expand the nodes of a proof tree, starting with its sole root obligation. Let S_i be the product of the weights of the current open obligations. Expanding a node replaces its conclusion factor by its two premise factors and gives the specified update. Truth obligations have factor one. At the end the accumulator is one. The proof does not require an ordered stack, decoding of a whole stack, or ranks on proof nodes.

Exact local polynomial. Let $H(p, q, v; \boldsymbol{z})$ be the branch-shared uniform-Horn polynomial in `horn_shared_source.json`. It is the product of twelve sums of squares, is nonnegative on all integer assignments, and uses seventeen natural local witnesses. Its exact degree is fifty. Set $M = M_0 + 1$ and introduce natural label slacks d_p, d_q, d_v. The local accumulator polynomial is exactly

$$\begin{aligned} A = {} & H(p, q, v; \boldsymbol{z}) \\ & + \big(S'(1 + v(M_0 + 1)) - S(1 + p(M_0 + 1))(1 + q(M_0 + 1))\big)^2 \\ & + (p + d_p - m)^2 + (q + d_q - m)^2 + (v + d_v - m)^2. \end{aligned}$$

Its nonnegativity means $A = 0$ enforces every displayed condition, including one genuine Horn rule. There are $17 + 3 + 3 = 23$ local witnesses: the Horn witnesses, three labels, and three slacks. The four exposed coordinates are S, S', m, M_0. The update has total degree five before squaring, so the whole polynomial still has degree fifty. This is a *local* count, not the variable count of the eventual consistency equation.

A closed, but uncompiled, constructor. Define $\beta(x, y, i) = x \bmod (1 + (i + 1)y)$. The five outer natural coordinates are t, x, y, m, M_0. The exact input predicate for the supplied constructor is

$$\begin{aligned} & \beta(x, y, 0) = M_0 + 2, \qquad \beta(x, y, t) = 1, \\ & \forall i < t \; \exists \boldsymbol{w} \in \mathbb{N}^{23} \quad A\big(\beta(x, y, i), \beta(x, y, i + 1), m, M_0; \boldsymbol{w}\big) = 0, \\ & \forall j < m \; \exists k \in \mathbb{N} \quad M_0 + 1 = (j + 1)k. \end{aligned}$$

The last line supplies the common-multiple property directly. If $m = 0$, all labels are zero and the positive accumulator cannot change, so the endpoints exclude this degenerate case. Similarly $t = 0$ is excluded. For completeness, select m, then $M = m!$, then a finite accumulator path, and finally CRT witnesses x, y. The two bounded blocks are converted by the existing bounded-forall constructor; all remaining coordinates are existentially closed; the direct quartic compiler is applied before, and without, the Jones finalizer.

`SharedAffineProductEquation.lean` specifies this entire constructor against the pinned APIs. It has not been elaborated or kernel checked, and its resulting ordinary polynomial has not been materialized or counted. The common multiple M, beta scale y, and the bounded-forall compiler's internal CRT/majorization scales are distinct witnesses. Identifying them could create a circular impossible bound, and is not part of this construction. The result therefore supplies a genuinely different source and a precise compilation target, not a claim that a smaller final consistency polynomial has already been obtained.

Checks and scope. The direct finite checks cover 340,878 target/flow cases, 166,650 pairwise-gcd cases, and 227,252 product/balance comparisons. The shared local polynomial has twelve independently re-expanded sum-of-squares factors and 480 modular local replays. Negative controls exhibit the dangers of multiple conclusions, signed multiplicities, insufficient radix, and missing coprimality. These are finite regressions, not replacements for the unbounded mathematical proofs or a Lean check.

9 A standalone asymmetric parity interface

This component bypasses the inflated central-binomial index, but does *not* yet constitute a new complete consistency equation. Its precise theorem concerns positive input integers S, T:

$$S \,\&\, T = 0 \quad \Longleftrightarrow \quad \exists \mathbf{w} \in \mathbb{N}^{12}\ P(S, T, \mathbf{w}) = 0.$$

The bit operation occurs only in this specification, not in the polynomial. The three-Pell quotient technique has a precedent in Jones–Sato–Wada–Wiens (1976), Section3.9. The specialization here uses odd-quotient proximity, rank locking, and a built-in congruence to remove one witness.

Write $n = S+T$ and use the twelve nonnegative witnesses $y, r, c, p, l, i, j, u_0, u_1, u_2, u_3, v$. Define

$$\begin{aligned}
x &= 2(y+1), & Y &= 2r+1,\\
M &= 128(n+1)(x+1)(Y+1), & A &= M(x+1),\\
C &= n+1+c, & K &= T+1+p(M-1),\\
L &= S+1+l(Mx-1), & D &= (A^2-1)C^2+1,\\
E &= 2(i+1)DC^2, & F &= (A^2-1)E^2+1,\\
G &= A+F(F-A), & H &= C+F\big(n+1+(2j+1)C\big),\\
I &= (G^2-1)H^2+1.
\end{aligned}$$

The five residuals are

$$\begin{aligned}
U_0 &= u_0^2 - \big((2n+6)^3(2n+8)(y+1)^2+1\big),\\
U_1 &= u_1^2 - \big((M^2-1)K^2+1\big),\\
U_2 &= u_2^2 - \big(((Mx)^2-1)L^2+1\big),\\
U_3 &= u_3^2 - DFI,\\
U_4 &= K^2L^2 - 64(C-YKL)^2 - v - 1.
\end{aligned}$$

Set $P = \sum_{a=0}^{4} U_a^2$. This is a literal ordinary polynomial of exact total degree 504, counting the two input coordinates as variables. The independent audit corrects an earlier draft's bound of 500.

A thirteen-witness, six-equation version retains a separate divisibility witness and has degree368. These are distinct tradeoffs.

Pell conventions and hypotheses. Let $\psi_a(k)$ be the second coordinate of $(a+\sqrt{a^2-1})^k$. For $a>1$, every nonnegative solution of $(a^2-1)z^2+1=\square$ is such a coordinate. Also $\psi_a(k)\equiv k \pmod{a-1}$, $(2a-1)^{k-1}\le\psi_a(k)\le(2a)^{k-1}$, and

$$(2a)^q\left(1-\frac{q}{2a}\right)\le\psi_a(q+1)\le(2a)^q.$$

The growth-square theorem applied to U_0 gives $x>128\cdot 2^n$. The indexed-Pell certificate of JSWW Lemma3.8 gives $C=\psi_A(n+1)$ from $C\ge n+1>1$, $DFI=\square$, and the prescribed two congruences of H. Here $F\equiv 1 \pmod{2C}$, so our formula for H supplies $H\equiv n+1 \pmod{2C}$ and $H\equiv C \pmod F$, with $H\ge n+1+2C$.

Soundness and rank locking. The Pell equations and congruences give

$$K=\psi_M(T+1+a(M-1)),\quad L=\psi_{Mx}(S+1+b(Mx-1)),\qquad a,b\ge 0.$$

The last residual implies $|C/(KL)-Y|<1/8$ and therefore $C/(KL)>7/8$. If $a>0$, the lower growth bound gives $K\ge(2M-1)^{2n+1}>2(2A)^n\ge 2C$, because $M-1\ge 2n+1$ and $(2M-1)^2>2A$. The same argument applies to L if $b>0$. Either case contradicts the proximity inequality. Thus both ranks are the intended ones.

Set $\rho=(x+1)^n/x^S$, $\sigma=C/(KL)$, and $\epsilon=n/(2M)$. The relative Pell bounds place both normalized numerator and denominator in $[1-\epsilon,1]$, whence $|\rho-\sigma|\le\sigma\epsilon/(1-\epsilon)<1/128$. The binomial expansion gives $\rho=Q+f$ with integer $Q\equiv\binom{n}{S} \pmod x$ and $0\le f<2^n/x<1/128$. Consequently $|Y-Q|<1/8+1/128+1/128<1$. Hence $Y=Q$ is odd. Since x is even, the binomial coefficient is odd. Lucas' binary criterion yields $S\,\&\,T=0$.

Completeness and the congruence compression. For disjoint positive S,T, choose a growth-square witness y and put $Y=\lfloor(x+1)^n/x^S\rfloor$, an odd positive integer. Define M,A,C,K,L at their intended Pell ranks. Their congruences supply nonnegative c,p,l. The indexed-Pell theorem supplies i and an initial coordinate H_{old} satisfying both congruences. Its coprimality identities separate D,F,I_{old} into squares, so H_{old} is a Pell coordinate for base G. The determinant-one Pell matrix is periodic modulo $2CF$. Increasing its index by multiples of a positive period preserves both congruences and gives arbitrarily large H_{new}. Choose it at least $C+F(n+1+C)$. Writing $H_{\text{new}}-C=Fq$ yields $q\equiv n+1+C \pmod{2C}$, hence $q=n+1+(2j+1)C$ with $j\ge 0$. This proves completeness of the compressed formula for H, rather than assuming an unproved bound for a particular canonical witness. The same relative estimates give $|\sigma-Y|<1/64<1/8$, so the last positive integer slack exists. The square-root witnesses exist by construction. This completes both directions of the theorem.

Exact degree and boundary. The successive degrees of M,A,C,D,E,F,G,H,I are $3,4,1,10,13,34,68,36,208$. Thus DFI has degree252 and U_3^2 degree504, with a nonzero leading term; no other squared residual cancels it. The theorem assumes $S,T>0$ and all twelve witnesses nonnegative. Substituting a packed source would still need a separate proved recovery of the binary base. That missing step is not hidden in this theorem, and no completed global improvement is claimed for this component.

10 Verification, provenance, and proof boundaries

Completed mathematical claims. The nearest-square lemma and the paired/nested witness transformations are proved above over all unrestricted integer assignments. The new Lucas slice

lemma treats negative parameters and the exceptional indices explicitly. Their composition with the pinned source theorem gives completed consistency equations. They are not merely bounded search results. The affine-flow theorem, disjunctive scratch reuse, and asymmetric parity theorem are also unbounded mathematical claims, but their new global consistency constructors are not finished.

Finite checks are evidence of different kinds. The archive checker first verifies its sealed SHA-256 file manifest and retained-member provenance. In a fresh temporary copy it executes only the newly authored builders and checkers. This prevents reproduction from modifying the delivered package. Exact generated outputs must rebuild byte for byte. Restricted AST readers are used for inherited polynomial strings; incoming Python is never imported. The independent source34 checker reconstructs the actual four compiler coefficients. The direct audit re-expands every local sum of squares and checks triangular, nonnegative constructor restoration. The displayed compact low-degree residuals have their own exact symbolic equality check. Finite signed, Pell-congruence, flow and modular-circuit examples are regressions, not substitutes for the unbounded arguments.

Negative controls. The release includes a manifest tamper test and mathematical mutation controls in the independent component audits. These have different meanings: rejecting changed bytes establishes packaging integrity, whereas rejecting a changed coefficient or invalid flow assumption exercises a mathematical check. Neither should be described as a formal kernel certificate. Optimized-Python runs are included for independent auditors using explicit exceptions, to make clear that those validations do not disappear under `python -O`.

Formal status reviewed at the pin. The latest repository work includes assembled source-to-ratio-to-finalizer bridges, a generic-ring evaluator, 31-to-16 elimination, and arithmetic source-equivalence work for an actual 89-record source. This progress must not be conflated with a certificate for a new literal circuit. The new packet's adapted formula identities and global endpoint still require the corresponding elaboration and proof-kernel replay. The pinned formal status documents are retained as inert evidence.

Why the fixed integers are not a loophole. The coefficient tuple is produced from a finite, explicitly specified proof-search source, not selected according to its unknown halting behavior. Payload hashes bind exact integers to exact source conventions. A large coefficient is counted as a fixed leaf in the operation ledger, not pretended to be free of specification cost. Raw and compressed sizes are separately reported. Different compiler-port conventions, especially shifted and native masks, are never silently interchanged.

Next decisive experiment. The most promising departure from the present finalizer is to elaborate `SharedAffineProductEquation.lean`, prove its local source bridge, then materialize and count the global polynomial produced by the direct bounded-forall compiler. That would test whether replacing ordered execution by commutative proof balance helps after compilation, rather than only at the local level. The common multiple, sequence CRT scale, and internal compiler majorants must remain separate until an explicit noncircular sharing theorem is proved. A local 17- or 23-witness count alone says nothing about the eventual nine- or ten-variable frontier.

A second experiment is integrating the asymmetric parity theorem with a low-cost binary-base guard. The missing binary-base implication is a specific obstacle, not a generic appeal to DPRM. Neither experiment is reported as a completed smaller equation here.

11 References and retained evidence

1. ProveIt, main commit `9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`, https://github.com/VladimirReshetnikov/ProveIt. The eight reviewed package identities, archive Git blobs and retained members are recorded in the accompanying source ledgers.

2. Z.-W. Sun, *Further results on Hilbert's Tenth Problem*, Theorems4.12–4.13, https://arxiv.org/html/1704.03504v7.

3. Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, modified Lucas certificate and interpolation, Theorems3.1 and4.1 and Propositions6.1–6.2, https://arxiv.org/html/2609.01594v1. The exact hypotheses used by the new slice argument are stated in Section 5; the audit provides a separate proof file.

4. J. P. Jones, D. Sato, H. Wada and D. Wiens, *Diophantine representation of the set of prime numbers*, American Mathematical Monthly83 (1976),449–464, especially Definition3.7, Lemma3.8 and Section3.9. https://www.cs.umd.edu/~gasarch/BLOGPAPERS/Jonesh10.pdf. The complete copyrighted article is not redistributed in this archive.

5. The inherited C222 numerical-calculus endpoint and constructive Diophantine compiler API are retained under `direct_audit/inherited/`, with their own earlier exact repository pin. These source snapshots expose the interfaces; their transitive imports are repository dependencies, not newly proved content.

6. Full auxiliary mathematical proofs appear in the component `PROOF.md`, `REVIEW.md`, `AUDIT.md`, and named theorem files. Read them together with the coefficient and source manifests.

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